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MID TERM EXAMINATION

March-2024

COURSE CODE- AI504

COURSE TITLE- ARTIFICIAL NEURAL NETWORK(ANN)

Time-1:30 Hours

Max. Marks-20

Note: Answer all questions. Assume suitable missing data, if any.

Q1. (a) Why do we use more than one layer in ANNs?

(b) Given an input feature vector of length 50, a fully connected feedforward neural network with one hidden layer containing 100 neurons, and an output layer with 3 neurons, how many total parameters are there in the network if each neuron has a bias term?

(c) How Backpropagation Through Time (BPTT) in Recurrent Neural Network(RNN) is different from conventional backpropagation of ANN.

(d) Consider an image with dimensions 200x200 pixels represented as a matrix. If we convolve this image with a filter/kernel of size 5x5 using no padding and a stride of 2, what will be the dimensions of the output feature map?

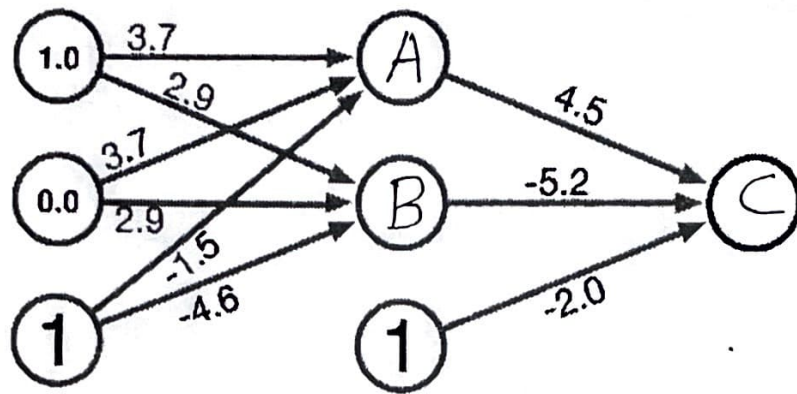
(e) You are working on a classification problem with a dataset containing 1000 samples. Explain how you would perform 5-fold cross-validation to evaluate a machine learning model.

(f) Consider a neural network model for sentiment analysis. You want to tune the learning rate and the number of hidden units in the hidden layer. Define a grid search space for these hyperparameters.

[2x6=12M]

Q2. Consider below ANN architecture with given input and weight values at input layer. Compute the value of neurons A and B at the hidden layer and the value of neuron C at the output layer. Assume the

value of bias associated with each node as 0 and activation function used as sigmoid.



[4 M]

Q3. You are training an Artificial Neural Network for anomaly detection in sensor data. The network has an input layer of size 50, two hidden layers with 80 and 60 neurons, respectively, and an output layer of size 1. The network uses the hyperbolic tangent (tanh) activation function in the hidden layers and the sigmoid activation function in the output layer. Calculate the total number of parameters in the network, including weights and biases.

[4 M]

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END TERM EXAMINATION

May-2024

COURSE CODE- AI504

COURSE TITLE- ARTIFICIAL NEURAL NETWORK(ANN)

Time-3:00 Hours

Max. Marks-40

Note: Answer all questions. Assume suitable missing data, if any.

Q1. (a) Compare and contrast LSTMs with traditional RNNs in terms of their ability to model long-term dependencies in sequences. [CO4]

(b) Explain the concept of vanishing gradients in deep neural networks. How does it affect training, and what are some techniques used to mitigate this problem? [CO3]

(c) Explain how back propagation can be used to solve the XOR problem which is not linearly separable ? [CO2]

(d) Suppose you have a multi-layer perceptron (MLP) with 100 input neurons, 50 hidden neurons, and 10 output neurons. If you increase the size of the training dataset from 1000 samples to 5000 samples, how does it impact the total number of parameters in the network? [CO1]

(e) Given a multi-layer perceptron (MLP) with an input layer of size 20, a hidden layer with 30 neurons, and an output layer with 10 neurons, calculate the total number of parameters in the network. Assume each neuron has a bias term. [CO1][$2 \times 5 = 10M$]

Q2. (a) Suppose you have a simple feedforward neural network with one input layer consisting of 2 neurons, one hidden layer consisting of 2 neurons, and one output neuron. The activation function is ReLU (Rectified Linear Unit) for hidden layers and sigmoid for the output layer. The initial weights from the input to the hidden layer are w_{11} ,

w_{12}], $[w_{21}, w_{22}]$ where $w_{11} = 0.5$, $w_{12} = -0.6$, $w_{21} = 0.1$, $w_{22} = 0.8$, and from the hidden to the output layer are $[w_{31}, w_{32}]$ where $w_{31} = -0.9$ and $w_{32} = 0.2$. Given an input vector $[0.7, 0.3]$, calculate the output of the network. Use a learning rate of 0.05 and perform backpropagation for one iteration using target output 0.6? [CO2] [6M]

(b) Consider an image with dimensions 400×400 pixels represented as a matrix. If we convolve this image with a filter/kernel of size 10×10 using no padding and a stride of 2, what will be the dimensions of the output feature map? [CO3] [4M]

Q3. (a) How does an LSTM based architecture help in solving the problem of gradient explosion in recurrent networks? Explain with the help of the gating structure of an LSTM cell. [CO4] [5M]

(b) Given an LSTM network with a single LSTM layer followed by a linear layer (fully connected) for classification, where the input size is 30, hidden size is 50, and output size is 10, calculate the total number of parameters in the LSTM network. [CO4] [5M]

Q4 (a) Describe the internal structure of an GRU unit and explain the purpose of each component (input gate, update gate, reset gate, candidate activation, final hidden state, and output gate). [CO5] [5M]

(b) Explain the concept of attention mechanisms in recurrent neural networks (RNNs). How do attention mechanisms improve the performance of RNNs on tasks involving long sequences, such as machine translation or text summarization? [CO4][5M]